An Improved Ellipse Propagation Model for Location Estimation in facilitating Ubiquitous Computing

Presenter : Junyang Zhou
Supervisor : Dr. Joseph Kee-yin Ng
Co-Supervisor: Dr. Jianliang Xu
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1. Introduction

Positioning Technology

Global Positioning System (GPS)

- A GPS receiver measures the distances from 3 or more satellites and utilize a mathematical method, called “Triangulation” to calculate the actual position

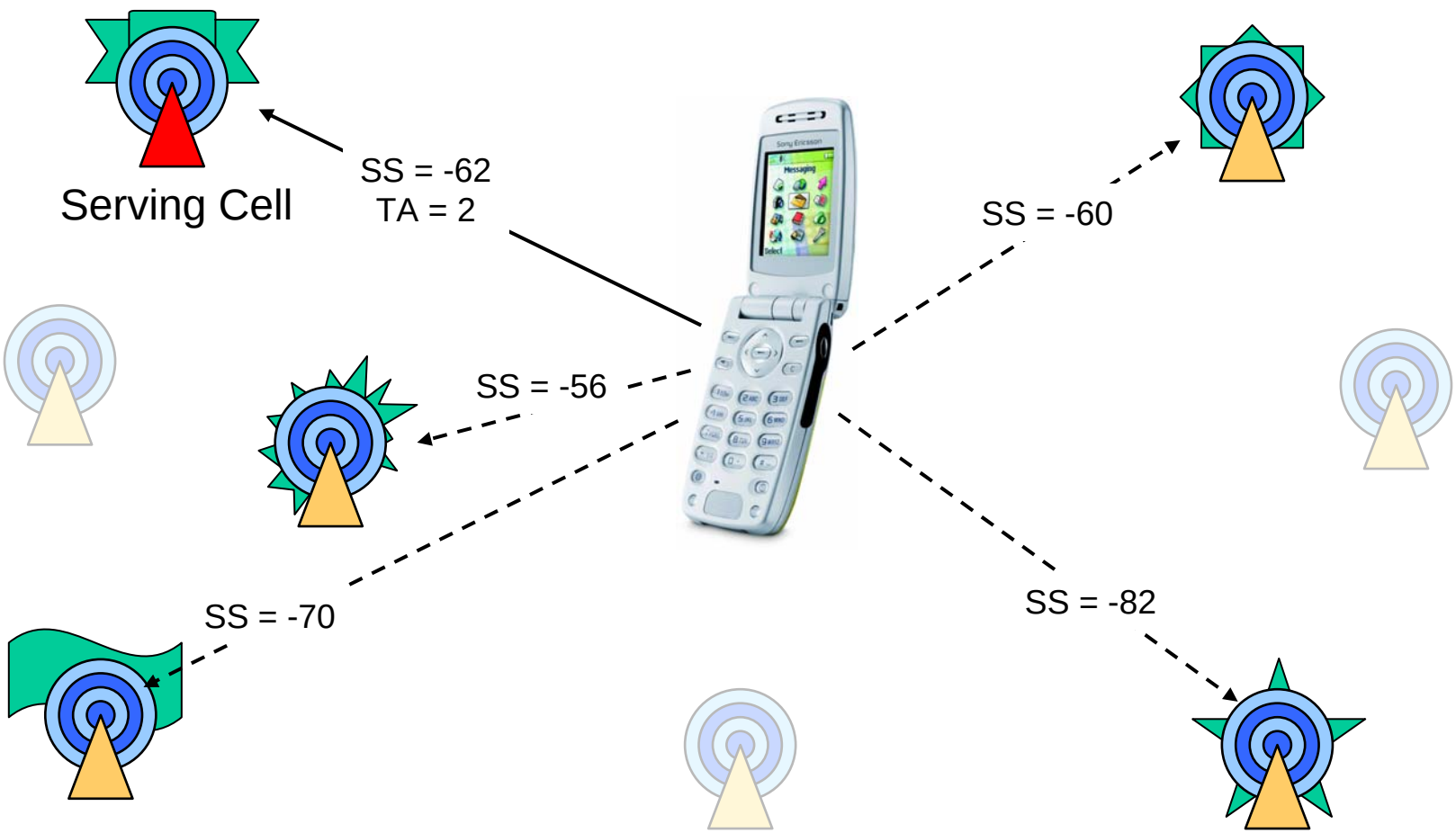
- Blind Spot Problem: Signals from the satellites would be blocked, and reflected by tall buildings.

Positioning Technology

Location Estimation based on Radio Cellular Phone Network

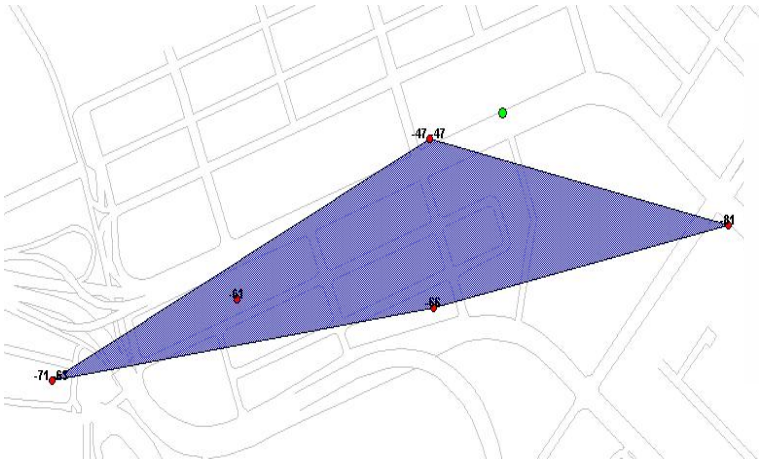
- CDMA: ToA, TDoA, E-OTD, AoA
- GSM: Signal Strength Based Algorithms
- Feasible for Metropolitan areas like Hong Kong

GSM Network



2. Related works

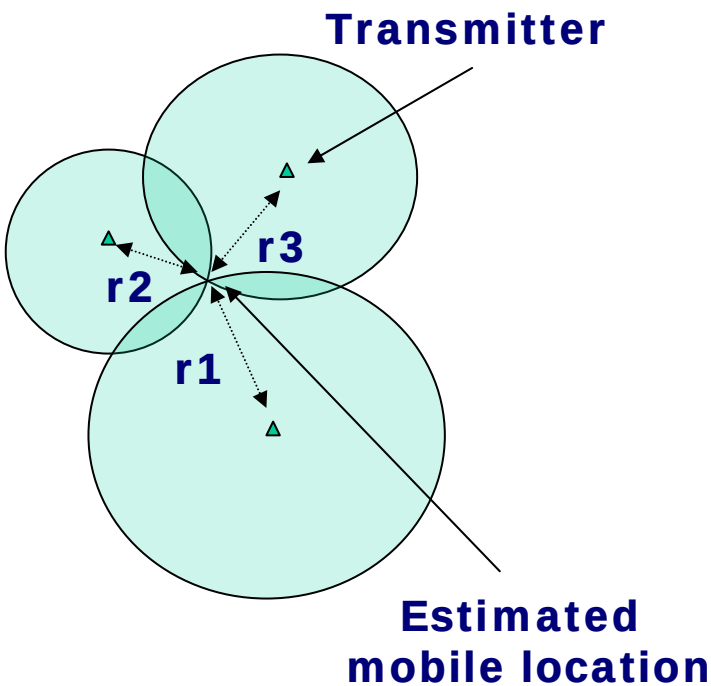
Center of Gravity Algorithm (CG)



$$\begin{cases} x = \dfrac{x_1 s_1^{-\alpha} + x_2 s_2^{-\alpha} + x_3 s_3^{-\alpha} + \ldots + x_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \\ y = \dfrac{y_1 s_1^{-\alpha} + y_2 s_2^{-\alpha} + y_3 s_3^{-\alpha} + \ldots + y_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \end{cases}$$

The “CG” estimation is always within the convex hull regardless the actual position of the MS is inside or outside this convex hull area

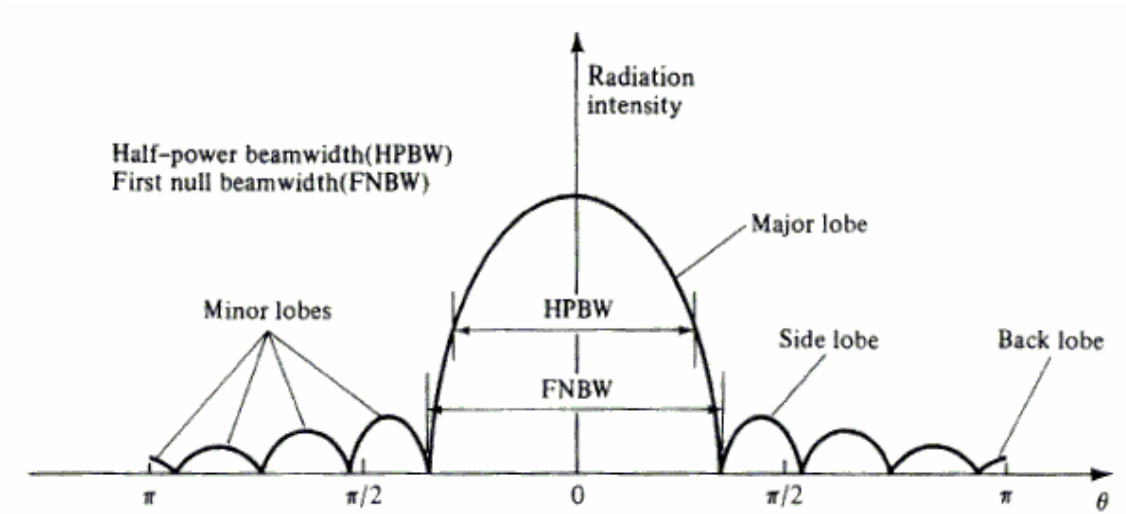
Circular Trilateration Algorithm (CT)

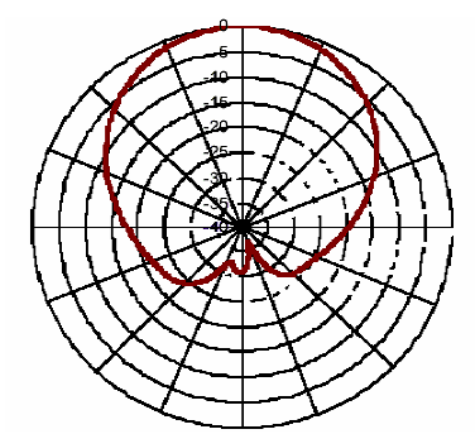


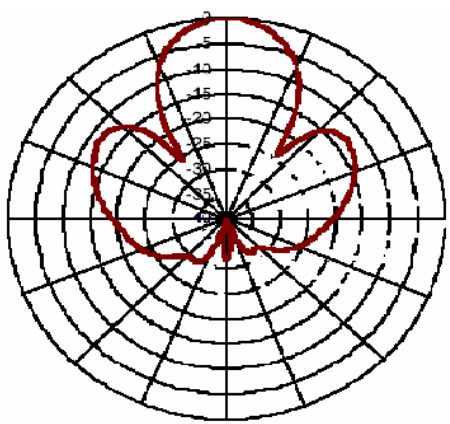
$$\begin{cases} (x-x_1)^2+(y-y_1)^2=(\frac{k}{s_1{}^{\alpha}})^2 \\ (x-x_2)^2+(y-y_2)^2=(\frac{k}{s_2{}^{\alpha}})^2 \\ (x-x_3)^2+(y-y_3)^2=(\frac{k}{s_3{}^{\alpha}})^2 \end{cases}$$

These three circles may not always intersect in the same point

The Directional Antenna







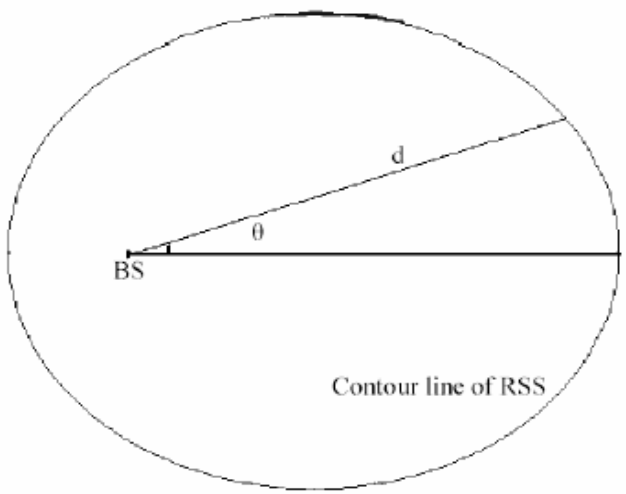
The Ellipse Propagation Model

Consider the contour line of the signal strength as an ellipse

$$d = k(s_0/s)^{1/\alpha}(1-e)/(1-e\cos(\theta))$$

where

d is the distance between MS and BS;
k is the proportion constant;
s_0 is the transmitting power of the BS;
s is the signal power received;
e is the eccentricity value of the ellipse, it describes the figure of the signal strength contour line;
θ is the deviation between the ellipse principal axis and the line of MS and BS;
α is called the path loss exponent.

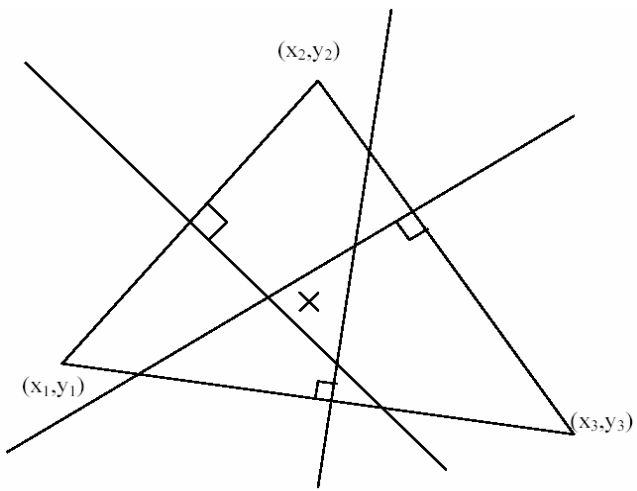


The Geometric Algorithm under EPM

Use three base stations not in a line to provide the MS location

$$\begin{cases} (x-x_1)^2+(y-y_1)^2=d_1^2 \\ (x-x_2)^2+(y-y_2)^2=d_2^2 \\ (x-x_3)^2+(y-y_3)^2=d_3^2 \end{cases}$$

Choose the “cross point” as the estimation of the Geometric Algorithm



3. The Iterative Algorithm under EPM

Motivation:

- One parameter of EPM (θ) is fixed in the Geometric Algorithm, errors may be induced as the surrounding changes
- The Iterative Algorithm is used to improve the Geometric Algorithm and to provide a more accurate and stable estimation

Idea:

- Use EPM to translate a received signal strength into a distance
- Consider the real location of the MS as a conditional expectation of the information of the received signal strengths and the BS locations
- Build up the iterative formula to provide the MS location
- Update the value of (x, y, θ) for each iteration
- Choose the convergence point as the estimation of the MS location

Structure of the Iterative Algorithm

Consider the "real" location is the conditional expectation

$$(x', y')^T = E((x, y)^T | x_0, y_0; s; l)$$

Rewrite it as two parts: one is certain, one is uncertain.

$$\begin{cases} x = f(x_0, y_0; s; l) + \epsilon \\ y = g(x_0, y_0; s; l) + \eta \end{cases}$$

Choose the certain terms as the estimation of the MS location

$$\begin{cases} x_0 = f(x_0, y_0; s; l) \\ y_0 = g(x_0, y_0; s; l) \end{cases}$$

Using Geometric Algorithm with Iterative Algorithm

Define the deviation θ as:

$$\theta_l = \begin{cases} \frac{5\pi}{2} - bear_l - \arccos\left(\frac{x-\alpha_l}{\sqrt{(x-\alpha_l)^2+(y-\beta_l)^2}}\right) & \text{if } y > \beta_l \\ \frac{\pi}{2} - bear_l + \arccos\left(\frac{x-\alpha_l}{\sqrt{(x-\alpha_l)^2+(y-\beta_l)^2}}\right) & \text{if } y \leq \beta_l \end{cases}$$

The iterative formulas under the Geometric Algorithm

$$\begin{aligned} x_{n+1} &= 2[(\beta_3-\beta_2)(d_1^2(n)-(\alpha_1^2+\beta_1^2)) \\ &\quad +(\beta_1-\beta_3)((d_2^2(n))-(\alpha_2^2+\beta_2^2)) \\ &\quad +\beta_2-\beta_1)(d_3^2(n)-(\alpha_3^2+\beta_3^2))]/|A| \\ y_{n+1} &= 2[(\alpha_2-\alpha_3)(d_1^2(n)-(\alpha_1^2+\beta_1^2)) \\ &\quad +(\alpha_3-\alpha_1)((d_2^2(n))-(\alpha_2^2+\beta_2^2)) \\ &\quad +(\alpha_1-\alpha_2)(d_3^2(n)-(\alpha_3^2+\beta_3^2))]/|A| \end{aligned}$$

The Iterative formulas can be rewritten as

$$\begin{cases} x_{n+1} = f(x_n, y_n; s_1, s_2, s_3; l_1, l_2, l_3) \\ y_{n+1} = g(x_n, y_n; s_1, s_2, s_3; l_1, l_2, l_3) \end{cases}$$

Convergence of the Iterative Algorithm

Define two distance functions

$$d(x, y, \alpha_l, \beta_l, p_{epm}, s_0, s) = \frac{k(s_0/s)^{1/\alpha}(1-e)}{1-e\cdot\cos(\theta(x,y,\alpha_l,\beta_l,bear_l))}$$

$$de(x, y, \alpha_l, \beta_l) = \sqrt{(x-\alpha_l)^2 + (y-\beta_l)^2}$$

The Convergence Theory

Theorem 3.1 *If* (α_1, β_1), (α_2, β_2), (α_3, β_3) *are not in the same line, and suppose that* $|\frac{d_i^2(x,y)}{de_i^2(x,y)} \cdot \frac{(y-\beta_i)}{de_i(x,y)} \cdot \frac{e_i \sin(\theta_i)}{1-e_i\cos(\theta_i)}| \leq 1$, $|\alpha_i - \alpha_j| \geq 4$, *if* $|\alpha_i - \alpha_j| \neq 0$; $|\beta_i - \beta_j| \geq 4$, *if* $|\beta_i - \beta_j| \neq 0$; *where* $i \neq j$ *and* $i, j = 1, 2, 3$. *Then* $\{x_n\}$ *and* $\{y_n\}$ *converge.*

The key points of the proof

Both $\{x_n\}$ and $\{y_n\}$ have bounds.

$(x_{n+1} - x_n) = K(x_n - x_{n-1})$, $|K| < 1$.

4. Simulation Results

Estimating the EPM parameters

The Lookup Table

Region	e_1	e_2	e_3	k	α	Region	e_1	e_2	e_3	k	α
Aberdeen	0.4	0	0	1.5	7.3	ShamShuiPo	0.4	0.6	0	0.55	8.4
CauseWayBay	0.4	0.9	0	1.45	4.5	ShaTin	0.7	0.7	0	1.45	6.1
Central	0	0.2	0	1.45	7.2	ShekKipMeiPark	0.6	0	0	1.5	6.6
CheungShaWan	0.1	0	0	1.25	8.4	SheungShui	0	0	0	1.5	8.1
FoTan	0.9	0	0	1.5	6.4	SheungWan	0.4	0.9	0	1.45	8.4
HappyValley	0.6	0	0	1.5	8.1	TaiKooShing	0.9	0	0	0.55	4.9
HungHom	0	0	0.1	1.45	7.5	TaiWai	0.3	0.4	0	0.55	5.6
KowloonBay	0.4	0.2	0.9	0.55	6.4	TaiWoHau	0.2	0	0	0.65	5.3
KowloonCity	0.6	0.3	0.5	1.5	6.7	TinShuiWai	0.7	0	0	0.55	5.6
KowloonTong	0	0	0	1.5	7.8	TsingYi	0.1	0.1	0	1.5	7.7
KwaiFong	0.9	0	0	0.55	6.1	TsuenWan	0.9	0.5	0	0.55	8.4
KwunTong	0.9	0	0	1.4	7.5	TszWanShan	0.4	0	0	1.45	7.6
LaiChiKok	0	0.9	0	0.55	6.9	TuenMun	0.3	0	0	1.5	7.3
LaiKing	0.7	0.9	0.7	0.55	5.7	WanChai	0.2	0	0	1.5	7.4
MaOnShan	0.8	0	0	0.55	8.4	WongTaiSin	0.9	0	0	1.5	7.7
Mongkok	0.6	0	0	1.5	6.7	YauTong	0	0	0	1.5	7.1
PE-MK	0.6	0.9	0	1.5	7.8	YauYatChuen	0.3	0	0	1.5	7.5
PrinceEdward	0.9	0.9	0.3	0.55	8.4	YuenLong	0.9	0	0.9	1.5	5.9

Results of the Iterative Algorithm

The Iterative Algorithm result and improvement over the Geometric Algorithm

Region	Ave.	Imp.	Std.	67%	90%	Region	Ave.	Imp.	Std.	67%	90%
Aberdeen	228.85	5.26%	135.40	226.30	483.61	ShamShuiPo	103.36	6.79%	55.17	119.61	193.02
CauseWayBay	258.62	28.42%	189.56	323.19	619.47	ShaTin	351.58	5.72%	172.12	401.61	577.97
Central	109.64	4.96%	64.37	143.96	201.92	ShekKipMeiPark	314.36	3.80%	170.07	379.47	570.76
CheungShaWan	120.47	12.45%	66.34	158.94	211.89	SheungShui	507.99	-9.37%	333.80	798.05	1007.03
FoTan	280.15	8.16%	134.60	356.50	460.99	SheungWan	116.10	7.25%	58.98	144.56	190.77
HappyValley	343.89	9.86%	222.31	430.83	714.78	TaiKooShing	250.84	36.26%	153.91	295.97	440.13
HungHom	934.48	-0.27%	528.76	1414.11	1838.55	TaiWai	226.03	3.20%	90.86	269.04	319.49
KowloonBay	199.82	14.3%	108.01	227.17	343.94	TaiWoHau	277.40	-6.97%	144.96	356.22	508.62
KowloonCity	223.78	10.06%	129.99	264.94	550.48	TinShuiWai	384.60	-0.29%	197.60	513.49	779.81
KowloonTong	255.83	13.45%	153.84	209.41	499.45	TsingYi	554.74	-13.17%	262.30	674.04	931.13
KwaiFong	182.95	3.21%	94.83	224.52	318.17	TsuenWan	143.42	8.78%	60.28	138.80	236.72
KwunTong	115.05	8.78%	58.84	126.32	202.43	TszWanShan	232.29	11.86%	135.04	274.57	448.45
LaiChiKok	325.22	2.34%	175.28	214.62	617.10	TuenMun	319.73	5.05%	153.89	421.09	516.47
LaiKing	682.62	14.58%	249.34	996.45	1181.29	WanChai	166.92	16.64%	112.18	182.60	334.99
MaOnShan	353.11	7.27%	116.73	410.16	500.64	WongTaiSin	306.57	-20.35%	147.16	377.40	515.72
Mongkok	88.36	3.49%	50.04	111.08	150.08	YauTong	387.53	5.18%	321.40	484.65	1030.33
PE-MK	110.08	4.68%	62.25	134.99	202.64	YauYatChuen	227.77	1.8%	113.69	278.87	389.61
PrinceEdward	118.29	14.51%	69.91	138.67	214.91	YuenLong	267.59	18.29%	285.83	229.26	976.63

Compare among CG, CT and Geometric Algorithm

Compare among the CG, CT, Geometric and Iterative Algorithm and its improvement

Model	Average Error	Improvement %	Std.	sample number	success ratio %
CG	495.76	27.08%	787.05	116354	96.93%
CT	470.99	23.24%	986.06	116354	76.28%
Geometric Algorithm	441.09	18.04%	721.12	116354	84.08 %
Iterative Algorithm	361.52	0%	541.87	116354	79.13%

5. Futrue work

- 1. Use the TA information to provide the MS location
- 2. Extend the EPM into a 3-D model
- 3. Find out a filter method to reduce the effect of the signal fluctuation

6. Q&A